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===== CodeSense C
- Complete Project Execution Guide
===== Project Name :
CodeSense C IDE : Visual Studio Code Compiler : GCC (MinGW) Platform : Windows
===== STEP 1 : Open
Visual Studio Code
===== 1. Open Visual
Studio Code. 2. Open Terminal Menu ---- Terminal ↓ New Terminal `
===== STEP 2 :
Create Project Folder
===== Go to E: Drive
E: Create Project Folder mkdir C_PROGRAMMING_PROJECTS Move into Project Folder cd
C_PROGRAMMING_PROJECTS Create Project mkdir CodeSense_C Move inside Project cd
CodeSense_C =====
STEP 3 : Create Project Structure
===== Create folders
mkdir include mkdir src mkdir samples mkdir reports mkdir docs
===== STEP 4 :
Create Source Files
===== Create Main
File type nul > main.c Create Source Files type nul > src\menu.c type nul > src\utils.c type nul >
src\file_reader.c type nul > src\tokenizer.c type nul > src\keyword_analyzer.c type nul >
src\identifier_analyzer.c type nul > src\operator_analyzer.c type nul > src\constant_analyzer.c
type nul > src\function_analyzer.c type nul > src\string_analyzer.c type nul >
src\memory_analyzer.c type nul > src\bracket_checker.c type nul > src\naming_checker.c type
nul > src\report_generator.c
===== STEP 5 :
Create Header Files
===== type nul >
include\analyzer.h type nul > include\utils.h type nul > include\tokenizer.h type nul >
include\report.h
===== STEP 6 :
Create Sample Files
===== type nul >
samples\sample1.c type nul > samples\sample2.c type nul > samples\sample3.c
===== STEP 7 :
Create Report File
===== type nul >
reports\analysis_report.txt
===== STEP 8 :
Create README File
===== type nul >
README.md =====
STEP 9 : Verify Folder Structure
===== Run dir
Expected Structure CodeSense_C | |— include |— src |— samples |— reports |—
docs | |— main.c |— README.md
===== STEP 10 :
Open Folder in VS Code
===== Run code . OR
File ↓ Open Folder ↓ Select CodeSense_C
===== STEP 11 :
Paste All Source Code
===== Open each file
and paste the code provided. Example main.c ↓ Paste main.c code ↓ Save Repeat for all files.
Shortcut Ctrl + S

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===== STEP 12 :
Compile Project
===== Open Terminal
Compile gcc main.c ^ src\menu.c ^ src\utils.c ^ src\file_reader.c ^ src\tokenizer.c ^
src\keyword_analyzer.c ^ src\identifier_analyzer.c ^ src\operator_analyzer.c ^
src\constant_analyzer.c ^ src\function_analyzer.c ^ src\string_analyzer.c ^
src\memory_analyzer.c ^ src\bracket_checker.c ^ src\naming_checker.c ^ src\report_generator.c
^ -linclude ^ -Wall ^ -Wextra ^ -std=c11 ^ -o CodeSense.exe OR (Single Line) gcc main.c
src\menu.c src\utils.c src\file_reader.c src\tokenizer.c src\keyword_analyzer.c
src\identifier_analyzer.c src\operator_analyzer.c src\constant_analyzer.c src\function_analyzer.c
src\string_analyzer.c src\memory_analyzer.c src\bracket_checker.c src\naming_checker.c
src\report_generator.c -linclude -Wall -Wextra -std=c11 -o CodeSense.exe
===== Compiler
Options ===== gcc →
GCC Compiler -linclude → Include Header Files -Wall → Enable All Common Warnings -Wextra
→ Enable Extra Warnings -std=c11 → Use C11 Standard -o → Output Executable Name
CodeSense.exe → Executable File
===== STEP 13 : Run
Project ===== Run
CodeSense.exe OR .\CodeSense.exe
===== STEP 14 :
Load Sample File
===== Menu 1. Load
C File Enter samples\sample1.c OR samples\sample2.c OR samples\sample3.c Press Enter
===== STEP 15 :
Perform Analysis
===== 1 Load File 2
Keyword Analysis 3 Identifier Analysis 4 Operator Analysis 5 Constant Analysis 6 Function
Analysis 7 String Analysis 8 Memory Analysis 9 Bracket Checking 10 Naming Convention
Checking 11 Complete Analysis 12 Generate Report 0 Exit
===== STEP 16 :
Generate Report
===== Choose 12
Report Generated reports\analysis_report.txt
===== STEP 17 : Exit
===== Choose 0
===== Common
Errors ===== 1. gcc is
not recognized Reason Compiler is not installed or not added to PATH. -----
----- 2. File does not exist Reason Wrong file path. Correct Path
samples\sample1.c ----- 3. Multiple Definition Error
Reason Same global variable declared in multiple files. Solution Use static or move the variable
to a single source file. ----- 4. Undefined Reference
Reason A source file is missing during compilation. Solution Compile all .c files together.
===== Complete
Workflow =====
Open VS Code ↓ Open Terminal ↓ Go to Project Folder ↓ Compile Project ↓ Run Executable ↓
Load Sample File ↓ Perform Analysis ↓ Generate Report ↓ Exit CodeSense_C/ | |— include/ |
|— analyzer.h | |— tokenizer.h | |— report.h | |— utils.h | |— src/ | |— menu.c |
|— utils.c | |— file_reader.c | |— tokenizer.c | |— keyword_analyzer.c | |—
identifier_analyzer.c | |— operator_analyzer.c | |— constant_analyzer.c | |—
function_analyzer.c | |— string_analyzer.c | |— memory_analyzer.c | |—
bracket_checker.c | |— naming_checker.c | |— report_generator.c | |— samples/ | |—
sample1.c | |— sample2.c | |— sample3.c | |— reports/ | |— docs/ | |— main.c
|— README.md
===== End of

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